11 Chemistry
Unit 2 Topic Test 2: Volumetric Analysis and Limiting Reactants

Details	
Student Name:	Teacher Name:
Date Set: ____/____/26	Due Date: ____/____/26

Curriculum Assessed

On completion of Unit 2 the student should be able to measure and analyse chemicals in water. This topic test assesses acid–base volumetric analysis (direct titrations), titration technique and error analysis, and limiting and excess reactants.

Task Overview

Section A – Multiple Choice: 10 questions, 1 mark each (10 marks). Answer on the separate multiple-choice answer sheet.
Section B – Short Answer: 4 questions (30 marks). Answer in this booklet.

Conditions

- Test conditions (revision topic test)
- VCE Chemistry Data Book provided
- Permitted materials: pens, pencils, highlighters, erasers, sharpeners, rulers and one scientific calculator (no CAS).
- Not permitted: notes of any kind, blank paper, white-out, mobile phones, smart watches or any other electronic device.
- 46 mins writing time (about 1.15 minutes per mark)

Marking

- 40 marks
- Marked according to teacher-made marking guide.
- Show all working in calculation questions; give final answers with units and an appropriate number of significant figures. Equations must be balanced and include states.

A+	A	B+	B	C+	C	D+	D	E+	E	NS
90-100%	80-89%	75-79%	70-74%	65-69%	60-64%	55-59%	50-54%	45-49%	40-44%	<40%

Total Marks:		Percentage:		Letter Grade:	

SECTION A – Multiple-choice questions (10 marks)

Answer **all** questions on the separate multiple-choice answer sheet by filling in the letter of the best response. Each question is worth 1 mark. No marks are deducted for incorrect answers. Do **not** write your Section A answers in this booklet.

Question 1

In a titration, the titre is

- **A.** the volume of the aliquot placed in the conical flask.
- **B.** the burette reading at the end point.
- **C.** the volume of solution delivered from the burette to reach the end point.
- **D.** the mean of all the titrations, including the rough titration.

Question 2

The amount of sodium hydroxide, NaOH, in 17.50 mL of 0.0800 M NaOH(aq) is

- **A.** 1.40×10^{-3} mol
- **B.** 1.40 mol
- **C.** 0.219 mol
- **D.** 4.57 mol

Question 3

When reading the volume in a burette, a student should

- **A.** read the top of the meniscus with their eye level with it.
- **B.** read the bottom of the meniscus while looking down from slightly above it.
- **C.** record the reading to the nearest 1 mL, because the scale is marked in whole millilitres.
- **D.** read the bottom of the meniscus with their eye level with it.

Question 4

Tartaric acid, $H_2C_4H_4O_6$, gives sour lollies their tang. A 25.00 mL aliquot of a tartaric acid solution was titrated with 0.112 M sodium hydroxide solution, NaOH(aq). The titre was 18.75 mL.

$$H_2C_4H_4O_6(aq) + 2NaOH(aq) \rightarrow Na_2C_4H_4O_6(aq) + 2H_2O(l)$$

The concentration of the tartaric acid solution is

- **A.** 0.0840 M
- **B.** 0.0420 M
- **C.** 0.168 M
- **D.** 0.0747 M

Question 5

Phosphorus, P_4, reacts with chlorine, Cl_2, to form phosphorus trichloride, PCl_3.

$$P_4(s) + 6Cl_2(g) \rightarrow 4PCl_3(l)$$

A mixture of 0.10 mol of P_4 and 0.50 mol of Cl_2 reacts until the reaction is complete. The maximum amount of PCl_3 that can form is

- **A.** 0.33 mol
- **B.** 0.40 mol
- **C.** 0.50 mol
- **D.** 0.75 mol

Use the following information to answer Questions 6 and 7.

Titanium metal, Ti, is used in aircraft frames and hip replacements. It is made by reacting titanium(IV) chloride, $TiCl_4$, with molten magnesium, Mg, at about 900 °C.

$$TiCl_4(g) + 2Mg(l) \rightarrow Ti(s) + 2MgCl_2(l)$$

In one batch, 760 kg of $TiCl_4$ reacts with 180 kg of Mg until the reaction is complete.

Question 6

Which row correctly identifies the limiting reactant and the mass of titanium formed?

	Limiting reactant	**Mass of Ti formed (kg)**
A.	$TiCl_4$	192
B.	Mg	355
C.	Mg	177
D.	Mg	88.7

Question 7

The mass of $TiCl_4$ remaining when the reaction is complete is closest to

A. 703 kg
B. 56.7 kg
C. 580 kg
D. 0.298 kg

Question 8

A rough titration is usually carried out before the accurate titrations.

The main purpose of the rough titration is

A. so that its titre can be included in the mean titre.
B. to check that the pipette delivers exactly its stated volume.
C. to measure the concentration of the solution in the burette.
D. to find the approximate titre, so that later titrations can be done carefully near the end point.

Question 9

A student pipetted aliquots of a hydrochloric acid solution, HCl(aq), into a conical flask and titrated them with a sodium hydroxide solution, NaOH(aq), of accurately known concentration from a burette.

Which one of the following would cause the calculated concentration of the hydrochloric acid to be lower than its true value?

A. The end point was overshot in each titration.
B. The pipette used delivered slightly more than its stated volume.
C. Some of the aliquot splashed out of the conical flask while it was being swirled.
D. At the end point, a drop of NaOH solution was left hanging from the burette tip and did not enter the flask.

Question 10

25.0 mL of 0.200 M nitric acid, $HNO_3(aq)$, is mixed with 20.0 mL of 0.300 M potassium hydroxide solution, KOH(aq).

$$HNO_3(aq) + KOH(aq) \rightarrow KNO_3(aq) + H_2O(l)$$

Which row correctly identifies the reactant in excess and the amount of it left over?

	Reactant in excess	Amount in excess (mol)
A.	KOH	1.00×10^{-3}
B.	HNO_3	1.00×10^{-3}
C.	KOH	6.00×10^{-3}
D.	KOH	0.100

END OF SECTION A

SECTION B – Short-answer questions (30 marks)

Answer **all** questions in the spaces provided. Show all working in calculation questions. Give final answers with units and appropriate significant figures. Equations must be balanced and include states.

Question 1 (9 marks)

Grace works in a soil-testing laboratory that extracts nutrients from soil samples with dilute hydrochloric acid, HCl(aq). Each batch of the acid must have a concentration between 0.175 M and 0.185 M.

To check a new batch, Grace titrated 25.00 mL aliquots of the acid with a standardised 0.165 M sodium hydroxide solution, NaOH(aq), delivered from a burette. The laboratory's rule is that a mean titre is calculated only from at least three concordant titres (titres within 0.10 mL of each other). The equation for the reaction is

$$NaOH(aq) + HCl(aq) \rightarrow NaCl(aq) + H_2O(l)$$

Titration	Rough	1	2	3
Final burette reading (mL)	29.40	29.25	30.25	29.55
Initial burette reading (mL)	0.10	0.55	1.20	0.80

a) Calculate the titre for titration 2.

1 mark

b) Explain why Grace cannot yet calculate a mean titre under the laboratory's rule, and state what she should do next.

2 marks

Grace did one more titration, which gave a titre of 28.80 mL. Her mean concordant titre was 28.75 mL.

c) Calculate the concentration of the hydrochloric acid, in mol L^{-1}, and state whether this batch meets the laboratory's requirement.

4 marks

d) To use less sodium hydroxide, a colleague suggests using 10.00 mL aliquots instead of 25.00 mL aliquots. Explain one disadvantage of this change.

2 marks

Question 2 (9 marks)

Urea, $CO(NH_2)_2$, is the most widely used nitrogen fertiliser in Australia. It is made by reacting ammonia, NH_3, with carbon dioxide, CO_2.

$$2NH_3(g) + CO_2(g) \rightarrow CO(NH_2)_2(s) + H_2O(l)$$

In a pilot plant, 340 kg of ammonia and 520 kg of carbon dioxide were mixed and the reaction went to completion.

a) Calculate the amount, in mol, of ammonia in 340 kg.

1 mark

b) Show by calculation that ammonia is the limiting reactant.

2 marks

c) Calculate the maximum mass of urea, in kg, that can form.

2 marks

d) Calculate the mass, in kg, of the excess reactant that remains.

2 marks

e) Explain why adding a small amount of extra ammonia to this mixture would increase the mass of urea formed, but adding extra carbon dioxide would not.

2 marks

Question 3 (6 marks)

Callum and Harriet each titrated 15.00 mL aliquots of the same nitric acid solution, $HNO_3(aq)$, with the same 0.0985 M sodium hydroxide solution, NaOH(aq). Their teacher, using careful technique, obtained a mean titre of 18.40 mL for the same acid. Each student's three titres after the rough titration are shown below.

Student	Titre 1 (mL)	Titre 2 (mL)	Titre 3 (mL)
Callum	18.95	18.90	18.95
Harriet	18.15	18.65	18.40

a) Identify the student whose titres are more precise.

1 mark

b) Callum then realised that, each time he filled his burette, he had left an air bubble in the burette tip below the tap, and the bubble was pushed out during the titration. Explain why this made his recorded titres larger than 18.40 mL.

2 marks

c) State whether Callum's air-bubble error is a random error or a systematic error.

1 mark

d) Callum suggests doing five more titrations with the same method and averaging all eight titres. Explain whether this would improve the accuracy of his result.

2 marks

Question 4 (6 marks)

Dev tested a vitamin C tablet labelled '500 mg ascorbic acid'. He crushed one tablet, dissolved it in about 50 mL of deionised water, H_2O, in a conical flask and titrated it with 0.105 M sodium hydroxide solution, NaOH(aq). The titre was 26.60 mL. Assume that only the ascorbic acid, $C_6H_8O_6$, in the tablet reacts with the NaOH.

$$C_6H_8O_6(aq) + NaOH(aq) \rightarrow NaC_6H_7O_6(aq) + H_2O(l)$$

a) Calculate the amount, in mol, of NaOH that reacted.

1 mark

b) Calculate the mass, in mg, of ascorbic acid in the tablet.

2 marks

c) Calculate the mass of ascorbic acid that Dev found as a percentage of the 500 mg stated on the label.

1 mark

d) Dev's teacher said that he would have obtained the same titre if he had dissolved the tablet in 80 mL of water instead of 50 mL. Explain whether the teacher is correct.

2 marks

END OF SECTION B

Additional space

Clearly number any response written here.

END OF TEST

11 Chemistry Unit 2 Topic Test 2: Volumetric Analysis and Limiting Reactants

SECTION A – MULTIPLE-CHOICE ANSWER SHEET

Student name: _____ Teacher: _____

Shade or circle ONE letter for each question. Each question is worth 1 mark. If you change an answer, cross out the first answer clearly.

Q	A	B	C	D
1	(A)	(B)	(C)	(D)
2	(A)	(B)	(C)	(D)
3	(A)	(B)	(C)	(D)
4	(A)	(B)	(C)	(D)
5	(A)	(B)	(C)	(D)
6	(A)	(B)	(C)	(D)
7	(A)	(B)	(C)	(D)
8	(A)	(B)	(C)	(D)
9	(A)	(B)	(C)	(D)
10	(A)	(B)	(C)	(D)

Section A score: _____ / 10